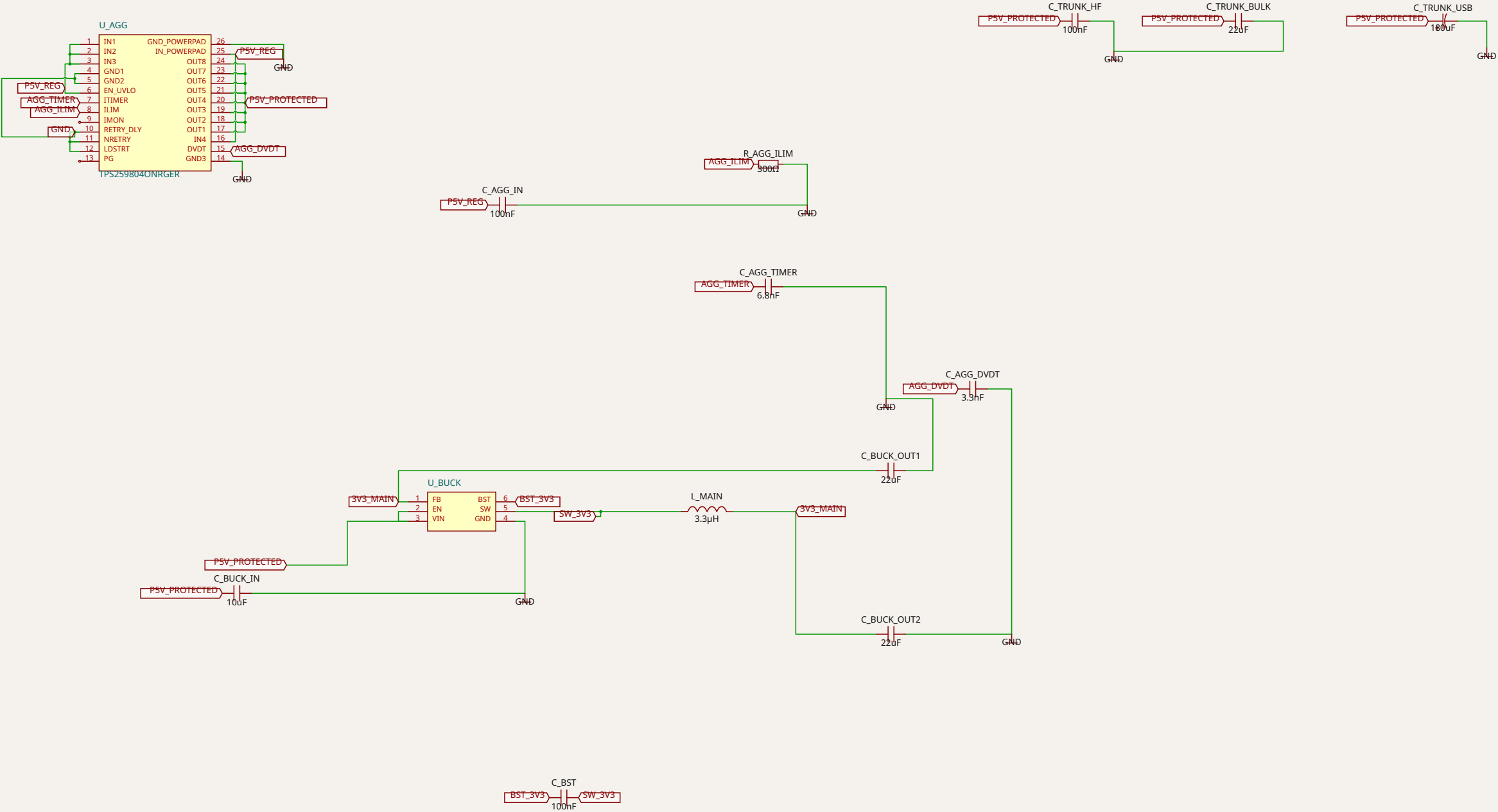


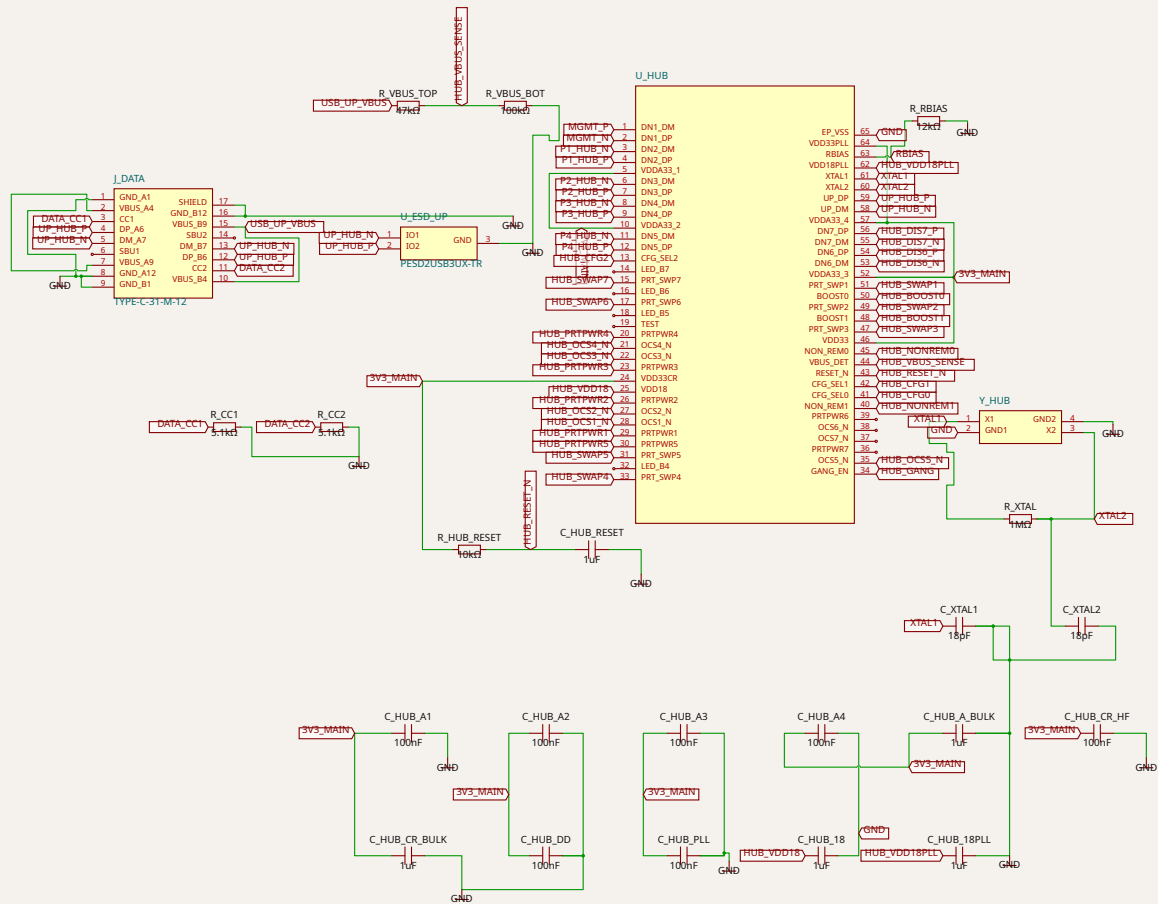
USB_CONTROLLED_DEBUG_HUB

POWER DISTRIBUTION — aggregate eFuse, USB bulk bank and 3.3 V buck



USB_CONTROLLED_DEBUG_HUB

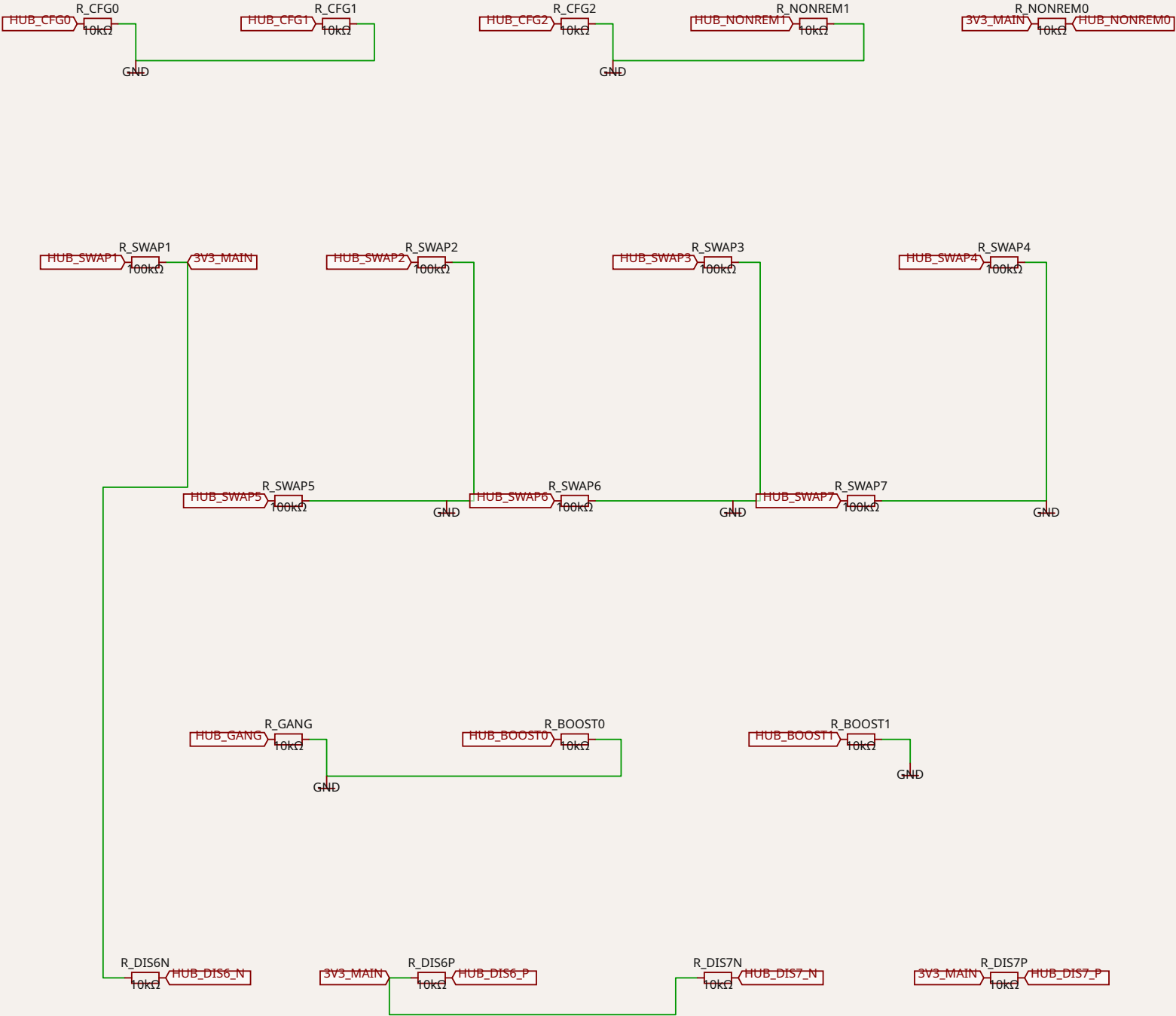
USB HUB — USB-C data-only upstream, ESD, USB2517I, straps, clock and bypass



INTENTIONAL NC: U_HUB LED/TEST and disabled-port 6/7 PWR/OCS pins.

USB_CONTROLLED_DEBUG_HUB

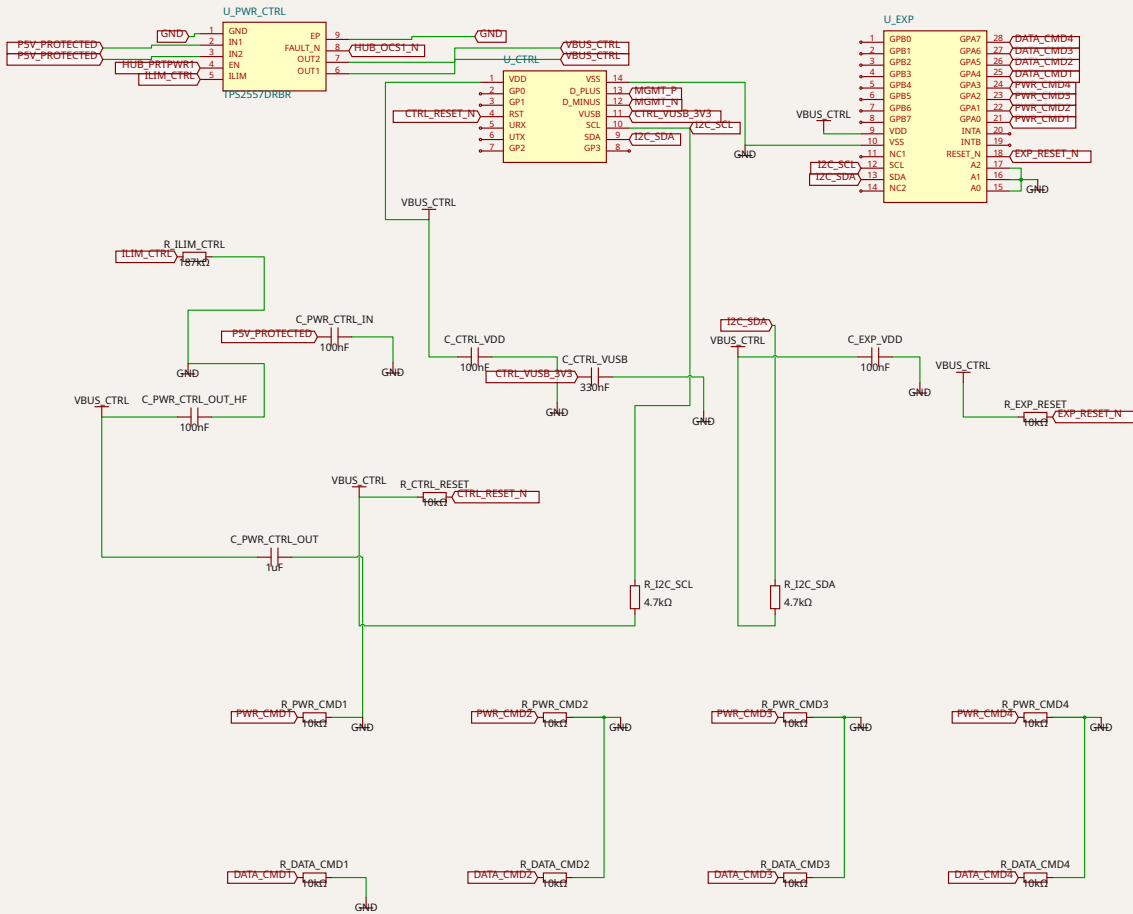
USB HUB CONFIGURATION — P1 swapped, P2-5 normal, P6-7 disabled



USB_CONTROLLED_DEBUG_HUB

MANAGEMENT — factory MCP2221A HID/I2C and MCP23017 command bank

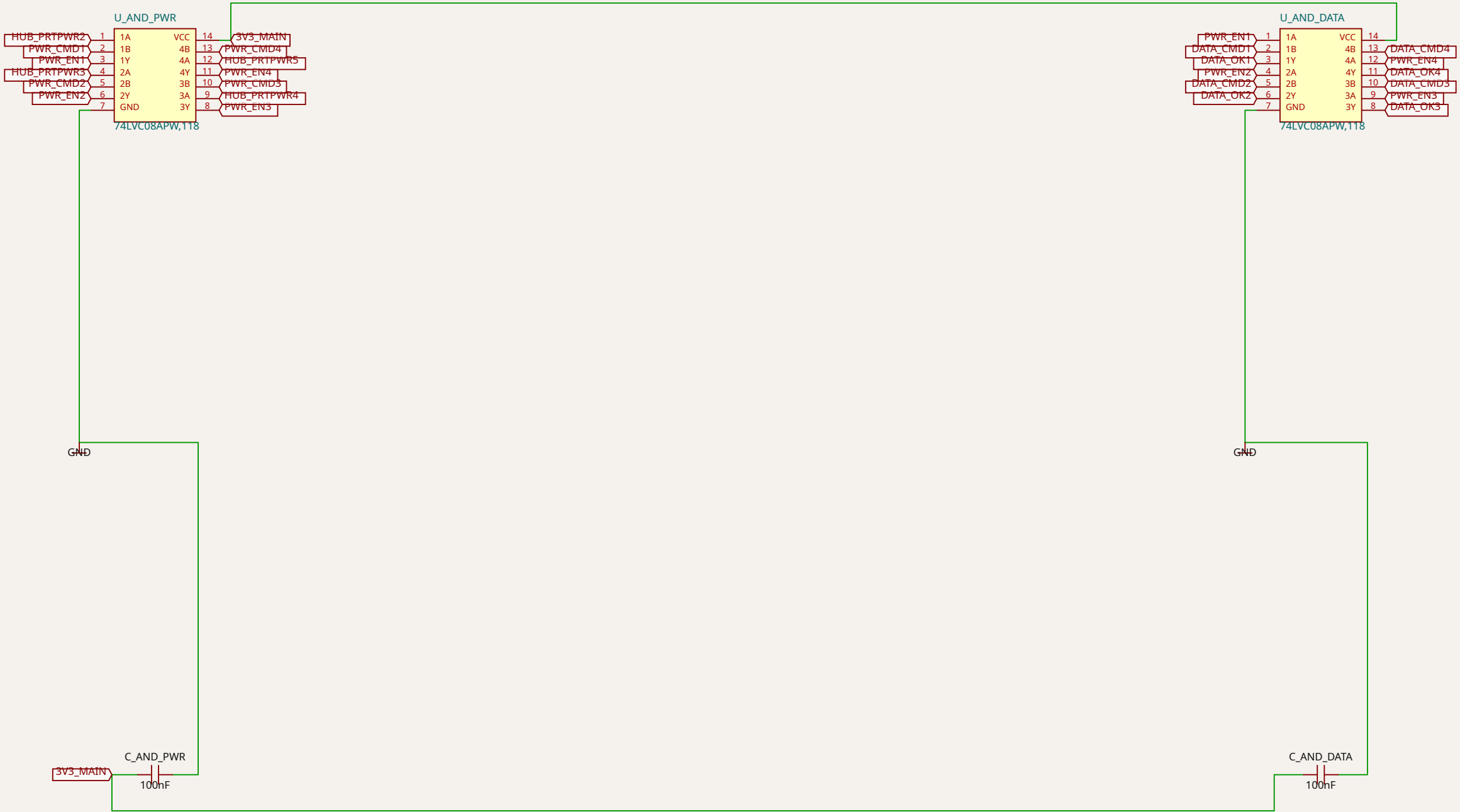
Page 5 of 10 • exact circuit.json SHA-256 e4a1bbc31c38d0a5... • 22 components • landscape page fit



INTENTIONAL NC: unused U_CTRL GPIO/UART; U_EXP GPB/INT/NC pins.

USB_CONTROLLED_DEBUG_HUB

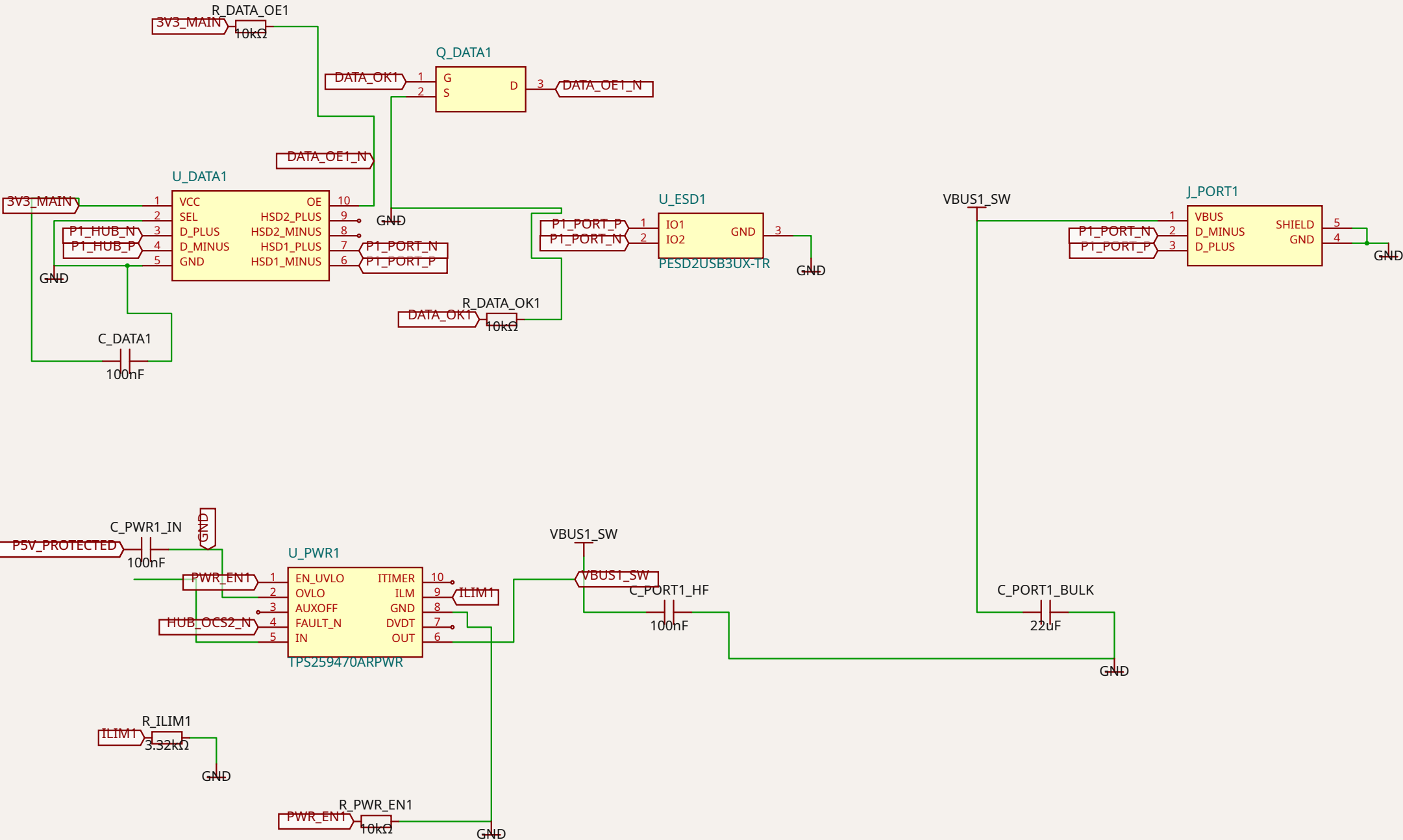
INTERLOCKS — hub policy AND host command; data follows commanded power enable



USB_CONTROLLED_DEBUG_HUB

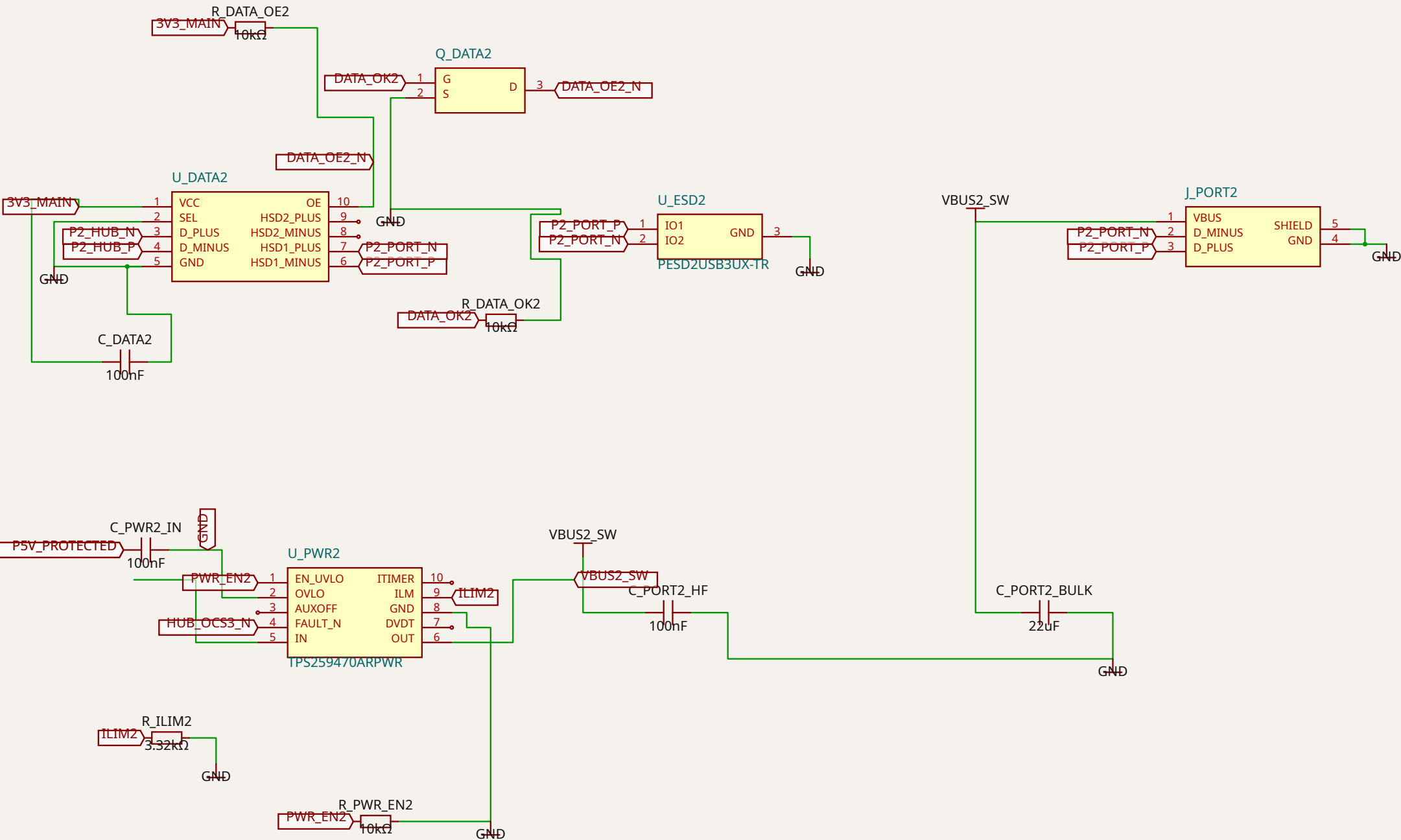
EXTERNAL PORT 1 — independent power/data disconnect

Page 7 of 10 • exact circuit.json SHA-256 e4a1bbc31c38d0a5... • 13 components • landscape page fit



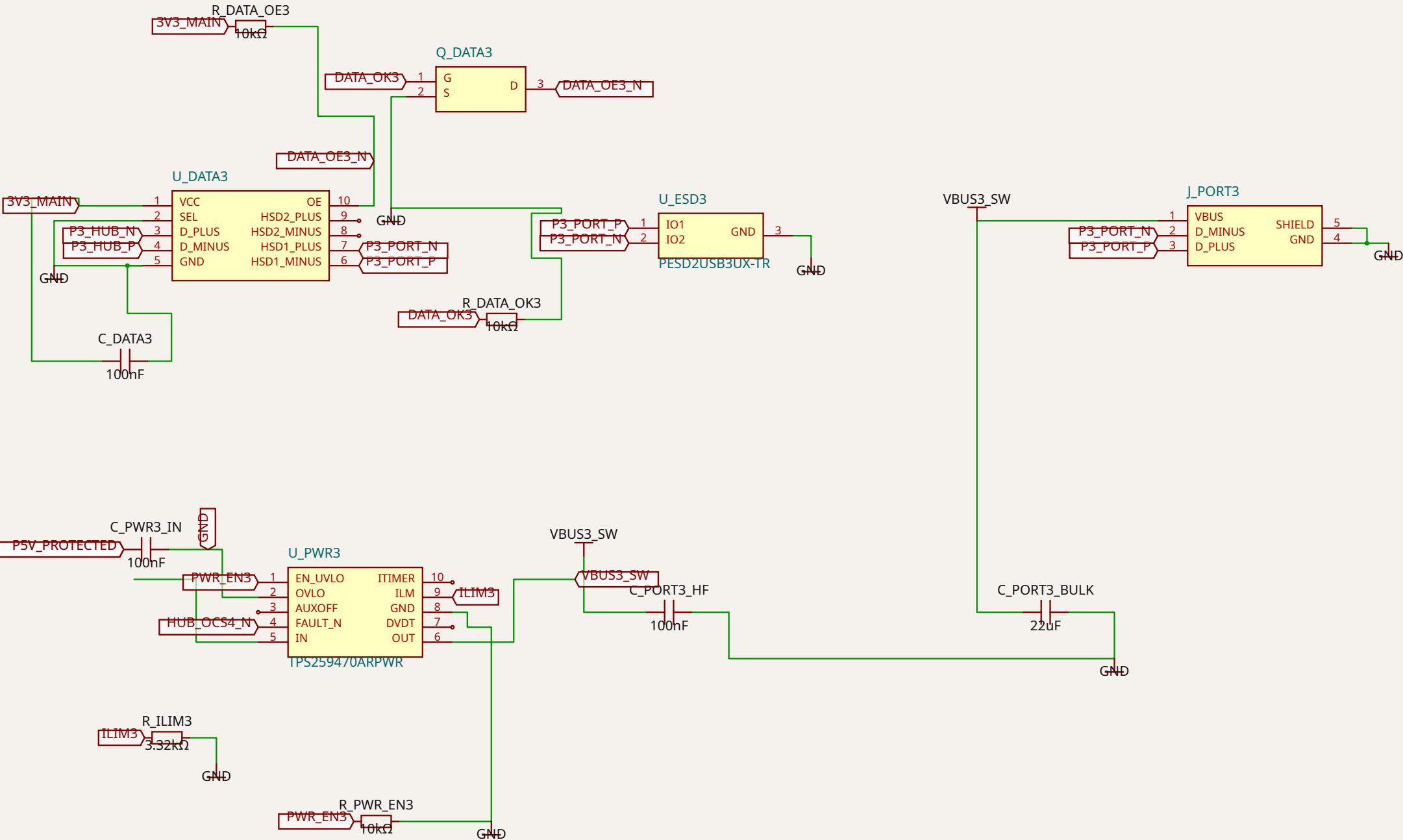
USB_CONTROLLED_DEBUG_HUB

EXTERNAL PORT 2 — independent power/data disconnect



USB_CONTROLLED_DEBUG_HUB

EXTERNAL PORT 3 — independent power/data disconnect



USB_CONTROLLED_DEBUG_HUB

EXTERNAL PORT 4 — independent power/data disconnect

